

```
R1 R2 R3 PC1 PC2
% Invalid input detected at '^' marker.

R1(config)#hostname 00010124
% Hostname contains one or more illegal characters.

00010124(config)#interface fastEthernet 0/0
00010124(config-if)#ip address 10.24.0.1 255.255.255.192
00010124(config-if)#no shutdown
00010124(config-if)#
*Mar 1 00:01:43.771: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:01:44.771: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
00010124(config-if)#exit
00010124(config)#interface fastEthernet 1/0
00010124(config-if)#ip address 10.24.0.65 255.255.255.224
00010124(config-if)#no shutdown
00010124(config-if)#
*Mar 1 00:02:31.635: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:02:32.635: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
00010124(config-if)#exit
00010124(config)#interface fastEthernet 0/1
00010124(config-if)#ip address 10.24.0.129 255.255.255.252
00010124(config-if)#no shutdown
00010124(config-if)#
*Mar 1 00:03:08.887: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:03:09.887: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
00010124(config-if)#exit
00010124(config)#ip route 10.24.0.2 255.255.255.192 10.24.0.128
%Inconsistent address and mask
00010124(config)#ip route 10.24.0.0 255.255.255.192 10.24.0.128
00010124(config)#ip route 10.24.0.65 255.255.255.224 10.24.0.28
%Inconsistent address and mask
00010124(config)#ip route 10.24.0.65 255.255.255.224 10.24.0.128
%Inconsistent address and mask
00010124(config)#ip route 10.24.0.64 255.255.255.224 10.24.0.128
00010124(config)#ip route 10.24.0.129 255.255.255.252 10.24.0.128
%Inconsistent address and mask

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R1 R2 R3 PC1 PC2
00010124(config)#interface fastEthernet 0/0
00010124(config-if)#ip address 10.24.0.130 255.255.255.224
00010124(config-if)#no shutdown
00010124(config-if)#
*Mar 1 00:07:48.855: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:07:49.855: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
00010124(config-if)#exit
00010124(config)#interface fastEthernet 1/0
00010124(config-if)#ip address 10.24.0.97 255.255.255.0
% 10.24.0.0 overlaps with FastEthernet0/0
00010124(config-if)#ip address 10.24.0.97 255.255.255.240
00010124(config-if)#no shutdown
00010124(config-if)#
*Mar 1 00:09:50.255: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:09:51.255: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
00010124(config-if)#exit
00010124(config)#interface fastEthernet 0/1
00010124(config-if)#ip address 10.24.0.133 255.255.255.224
% 10.24.0.128 overlaps with FastEthernet0/0
00010124(config-if)#show ip
^
% Invalid input detected at '^' marker.

00010124(config-if)#end
00010124#sho
*Mar 1 00:15:25.867: %SYS-5-CONFIG_I: Configured from console by console
00010124#show ip
% Incomplete command.

00010124#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.24.0.130 YES manual up up
FastEthernet0/1 10.24.0.133 YES manual administratively down down
FastEthernet1/0 10.24.0.97 YES manual up up
FastEthernet2/0 unassigned YES unset administratively down down
00010124#

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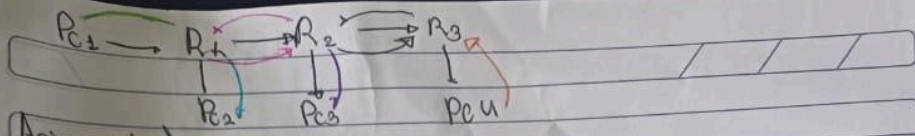
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R1 R2 R3 PC1 PC2
*Mar 1 00:00:03.995: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
*Mar 1 00:00:04.035: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to down
*Mar 1 00:00:04.035: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down
R2#enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#hostname 00010124
% Hostname contains one or more illegal characters.

00010124(config)#interface fastEthernet 0/0
00010124(config-if)#ip address 10.24.0.130 255.255.255.224
00010124(config-if)#no shutdown
00010124(config-if)#
*Mar 1 00:07:48.855: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:07:49.855: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
00010124(config-if)#exit
00010124(config)#interface fastEthernet 1/0
00010124(config-if)#ip address 10.24.0.97 255.255.255.0
% 10.24.0.0 overlaps with FastEthernet0/0
00010124(config-if)#ip address 10.24.0.97 255.255.255.240
00010124(config-if)#no shutdown
00010124(config-if)#
*Mar 1 00:09:50.255: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:09:51.255: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
00010124(config-if)#exit
00010124(config)#interface fastEthernet 0/1
00010124(config-if)#ip address 10.24.0.133 255.255.255.224
% 10.24.0.128 overlaps with FastEthernet0/0
00010124(config-if)#show ip
^
% Invalid input detected at '^' marker.

00010124(config-if)#end
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R1 R2 R3 PC1 PC2
*Mar 1 00:00:03.983: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down
*Mar 1 00:00:03.995: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
*Mar 1 00:00:04.035: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to down
*Mar 1 00:00:04.039: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down
R3#enable
R3#terminal config
^
% Invalid input detected at '^' marker.

R3#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface fastEthernet 0/0
R3(config-if)#ip address 10.24.0.134 255.255.255.224
R3(config-if)#no shutdown
R3(config-if)#
*Mar 1 00:12:36.955: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:12:37.955: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config-if)#interface fastEthernet 1/0
R3(config-if)#ip address 10.24.0.113 255.255.255.240
R3(config-if)#no shutdown
R3(config-if)#
*Mar 1 00:13:33.699: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:13:34.699: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R3(config-if)#exit
R3(config)#ip route 10.24.0.112 255.255.255.240 10.24.0.132
R3(config)#ip route 10.24.0.130 255.255.255.240 10.24.0.132
%Inconsistent address and mask
R3(config)#ip route 10.24.0.131 255.255.255.240 10.24.0.132
%Inconsistent address and mask
R3(config)#ip route 10.24.0.130 255.255.255.224 10.24.0.132
%Inconsistent address and mask
R3(config)#
```



Asignación de IP:

R1	Fa 0/0	10.24.0.1/26	...	192	PC1 → R1
R1	Fa 0/0	10.24.0.25/27	...	254	PC2 → R1
R1	Fa 0/1	10.24.0.19/30	...	252	R1 → R2
R2	Fa 0/0	10.24.0.130/30	...	252	R1 → R2
R2	Fa 0/0	10.24.0.97/24	...	240	R3 → R2
R3	Fa 0/1	10.24.0.133/30	...	252	R2 → R3
R3	Fa 0/0	10.24.0.134/30	...	252	R2 → R3
R3	Fa 0/0	10.24.0.113/24	...	240	PC4 → R3

PC1: 10.24.0.1
 PC2: 10.24.0.64
 PC3: 10.24.0.98
 PC4: 10.24.0.112

R1: 10.24.0.1
 PC1: 10.24.0.2

R1: 10.24.0.6
 PC2: 66

00010124

00 01 01 /24

10.24.0.0 /24

Salto = 64

Lea admin 45 host $2^n - 2 \geq 45$ $n = 8$ /26

10.24.0.0/26 1^a ip: 10.24.0.1 Última 10.24.0.62 10.24.0.63

Vendas 20 host $n = 5$ Salto 32 $32 - 5 =$ /27

10.24.0.64 1^a ip 10.24.0.65 Última 10.24.0.94 10.24.0.95

Sapoete 24 host $n = 4$ Salto: 16 /28

10.24.0.96 1^a ip 10.24.0.97 Última: 10.24.0.110 10.24.0.111

Gerência 20 host $n = 4$ Salto 16 /28

10.24.0.112 10.24.0.113 10.24.0.126 10.24.0.127

Enlace R₁-R₂ 2 host $n = 2$ Salto = 4 /30 = 224

10.24.0.128 10.24.0.129 10.24.0.130 10.24.0.131

Enlace R₂-R₃ 2 host $n = 2$ Salto = 4 /30 = 224

10.24.0.132 10.24.0.133 10.24.0.134 10.24.0.135

VS LM | range sub | Broadcast

Admin PC 10.24.0.1/26

~~PC1~~ 10.24.0.62/26 10.24.0.1/26

Vendas 10.24.0.65/27

PC2 10.24.0.68/27 10.24.0.95/27

Sapoete 10.24.0.97/24

PC3 10.24.0.98/24 10.24.0.117/24

Gerência 10.24.0.113/24

PC4 10.24.0.114/26 10.24.0.127/24

R₁ → R₂ 10.24.0.129

130

10.24.0.131/30

R₂ → R₃ 10.24.0.133

134

.135/30

192.24.0.0

0-0-0-0
000 10124

PC1: 40 host	n=6 / 26	Salto: 64		
192.24.0.0	192.24.0.7	192.24.0.62	192.24.0.63	
PC2: 28 host	n=5 / 27	Salto: 92		
192.24.0.64	192.24.0.65	192.24.0.94	192.24.0.95	
PC3: 18 host	n=5 / 27	Salto: 32		
192.24.0.96	192.24.0.97	192.24.0.126	192.24.0.127	
PC4: 9 host	n=4 / 28	Salto: 16		
192.24.0.128	192.24.0.129	192.24.0.142	192.24.0.143	
PC5: 5 host	n=3 / 29	Salto: 8		
192.24.0.144	192.24.0.145	192.24.0.146	192.24.0.147	
R1 → R2	n=2 / 30	Salto: 4		
192.24.0.148	192.24.0.149	150	151	
R1 → R3	n=2 / 30	Salto: 4		
152	153	154	155	
R2 → R4	n=2 / 30	4		
156	157	158	159	
R3 → R4	n=2 / 30	4		
160	161	162	163	
R4 → R5	n=2 / 30	4		
164	165	166	167	

